

Contact-Model-Aware Selective Passivation

A Proof-of-Concept Study

1 Proof-of-concept setup

A planar two-link robot presses a spherical pad against a horizontal surface while tracking a normal force of $F_n^d = 5$ N and a tangential velocity of $v_t^d = 0.05$ m/s. Control runs at 1 kHz, and the nominal surface friction is $\mu \approx 0.30$. Scripted helping and blocking forces act at the end effector.

The measured interaction is separated into a predicted task-contact wrench, a model-error residual, and the known human wrench. The controller maintains separate residual and human energy ledgers and filters the two-dimensional tangential-normal command when a power or energy bound becomes active. The human force is assumed known in this study; the objective is to evaluate the control architecture independently of sensing uncertainty.

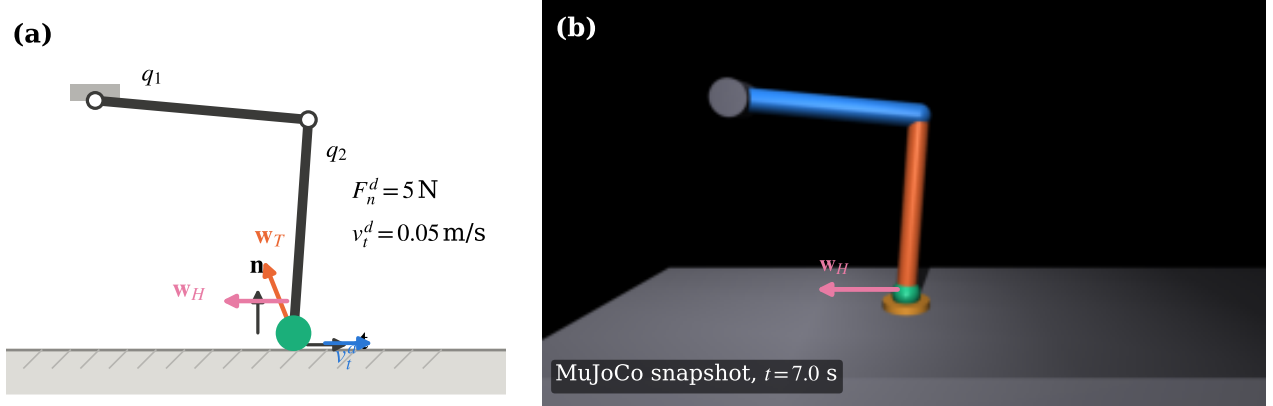


Figure 1: Proof-of-concept task. (a) Planar abstraction with the task-contact and human wrenches. (b) MuJoCo visualization at $t = 7.0$ s during the 3-N blocking interaction.

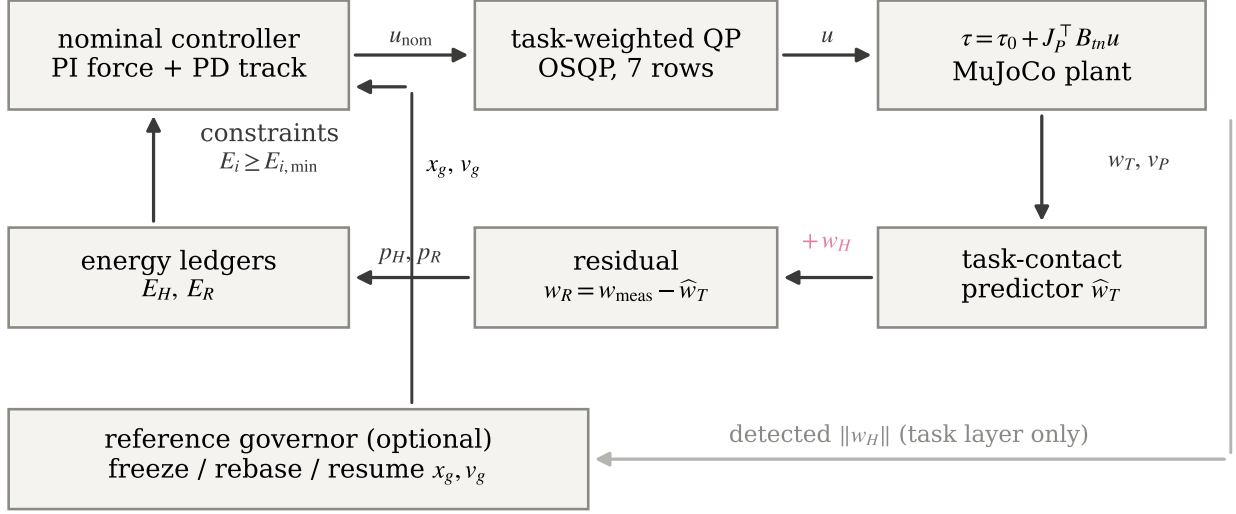


Figure 2: Controller architecture. The contact model removes predictable task wrench from the measured interaction. Residual and human power feed separate energy ledgers, whose limits constrain the task-weighted command filter. The reference governor, enabled only in C4, changes the nominal tangential reference but not the passivity certificate.

2 Proof-of-concept results

Five configurations isolate the main contributions (Table 1). C0 is the nominal force–motion controller. C1 applies passivity accounting to the complete measured contact port. C2 instead accounts only for contact that the task model does not explain. C3 adds a separate human-energy ledger and human-power limit, with no reference governor. C4 is the same dual-ledger QP with the reference governor enabled. Thus, C3 and C4 have identical passivity constraints; only C4 governs the nominal motion reference.

Table 1: Controller configurations used in the proof of concept.

Mode	Accounted power	Protected energy	Governor	Purpose
C0	none	none	off	nominal baseline
C1	measured contact	E_W	off	whole-port baseline
C2	residual	E_R	off	model-aware accounting
C3	residual and human	E_R, E_H	off	dual-ledger QP
C4	residual and human	E_R, E_H	on	governed C3

Selective-passivation filter

The measured interaction is decomposed into model mismatch and residual wrenches,

$$\mathbf{w}_\Delta = \mathbf{w}_T - \hat{\mathbf{w}}_T, \quad \mathbf{w}_R = \mathbf{w}_{\text{meas}} - \hat{\mathbf{w}}_T = \mathbf{w}_H + \mathbf{w}_\Delta, \quad (1)$$

$$p_H = \mathbf{w}_H^\top \mathbf{v}_P, \quad p_R = \mathbf{w}_R^\top \mathbf{v}_P = p_H + p_\Delta. \quad (2)$$

Here $p_\Delta = \mathbf{w}_\Delta^\top \mathbf{v}_P$; $p_i > 0$ denotes energy entering the robot, whereas $p_i < 0$ denotes energy delivered by the robot through port i . Each ledger is updated from realized power as

$$E_{i,k+1} = \min(E_{i,\max}, E_{i,k} + \Delta t p_{i,k}^{\text{actual}}), \quad i \in \{H, R\}. \quad (3)$$

Positive power charges the ledger up to its cap, while negative power discharges it. The logged raw energy is not lower-clamped, so any floor violation remains observable.

In the tangential–normal plane, the one-step velocity prediction and robust lower port power are

$$\mathbf{v}_{k+1}^{\text{pred}}(\mathbf{u}) = \mathbf{A}_k \mathbf{u} + \mathbf{b}_k, \quad \underline{p}_i(\mathbf{u}) = \mathbf{w}_i^\top \mathbf{v}_{k+1}^{\text{pred}}(\mathbf{u}) - \|\mathbf{w}_i\| \bar{e}_v, \quad (4)$$

with

$$\bar{e}_v = 0.025 \text{ m/s}.$$

Under the configured error bound, $\underline{p}_i$ conservatively lower-bounds realized port power.

For $\mathbf{u} = [u_t, u_n]^\top$, C3 solves the task-weighted QP

$$\begin{aligned} \mathbf{u}^* &= \arg \min_{\mathbf{u}} \frac{1}{2} (\mathbf{u} - \mathbf{u}_{\text{nom}})^\top \mathbf{W} (\mathbf{u} - \mathbf{u}_{\text{nom}}) + \epsilon_u \|\mathbf{u}\|^2, \\ \text{s.t.} \quad & E_i + \Delta t \underline{p}_i(\mathbf{u}) \geq E_{i,\text{safe}}, \\ & \underline{p}_i(\mathbf{u}) \geq -k_{\text{cbf}}(E_i - E_{i,\text{safe}}), \quad i \in \{H, R\}, \\ & -p_H^{\text{pred}}(\mathbf{u}) \leq P_H^{\max}, \quad \text{plus torque and torque-rate bounds.} \end{aligned} \quad (5)$$

The implemented parameters are

$$\begin{aligned} \mathbf{W} &= \text{diag}(1, 20), \quad k_{\text{cbf}} = 10 \text{ s}^{-1}, \quad P_H^{\max} = 0.10 \text{ W}, \\ E_{i,\text{safe}} &= E_{i,\min} + 10^{-4} \text{ J}. \end{aligned}$$

The larger normal weight favors tangential changes and preserves the normal-force command. The certificate covers only the residual and human ports: model-consistent task work is intentionally exempted, so global passivity of the complete actively controlled robot is not claimed.

Across the reported runs, the largest prediction error was 0.0115 m/s, below $\bar{e}_v$; median QP solution time was 24–44 μs , compatible with the 1 kHz control rate.

2.1 Model-aware accounting preserves useful task work

Whole-port passivity cannot distinguish useful contact work from an unsafe interaction. With an exact contact model and no human, ordinary sliding friction drains C1’s energy ledger and eventually stops the motion (Fig. 3). C3 subtracts the predicted task-contact wrench, leaving zero residual in this case. It consequently matches the nominal controller: $\text{RMSE}(F_n) = 0.033 \text{ N}$ and $\text{RMSE}(v_t) = 0.0025 \text{ m/s}$, with less than 0.02 % degradation and no ledger depletion.

Model error is not ignored; it appears in the residual channel. When the friction estimate is changed from the true value $\mu \approx 0.30$ to $\hat{\mu} = 0.20$, the residual ledger falls from 0.30 to 0.023 J, whereas the human ledger stays at 0.05 J (Fig. 4). The first passivity intervention is delayed from 5.95 s with whole-port accounting to 14.59 s with residual accounting. Thus, the controller responds to unexplained power rather than to all dissipative task contact.

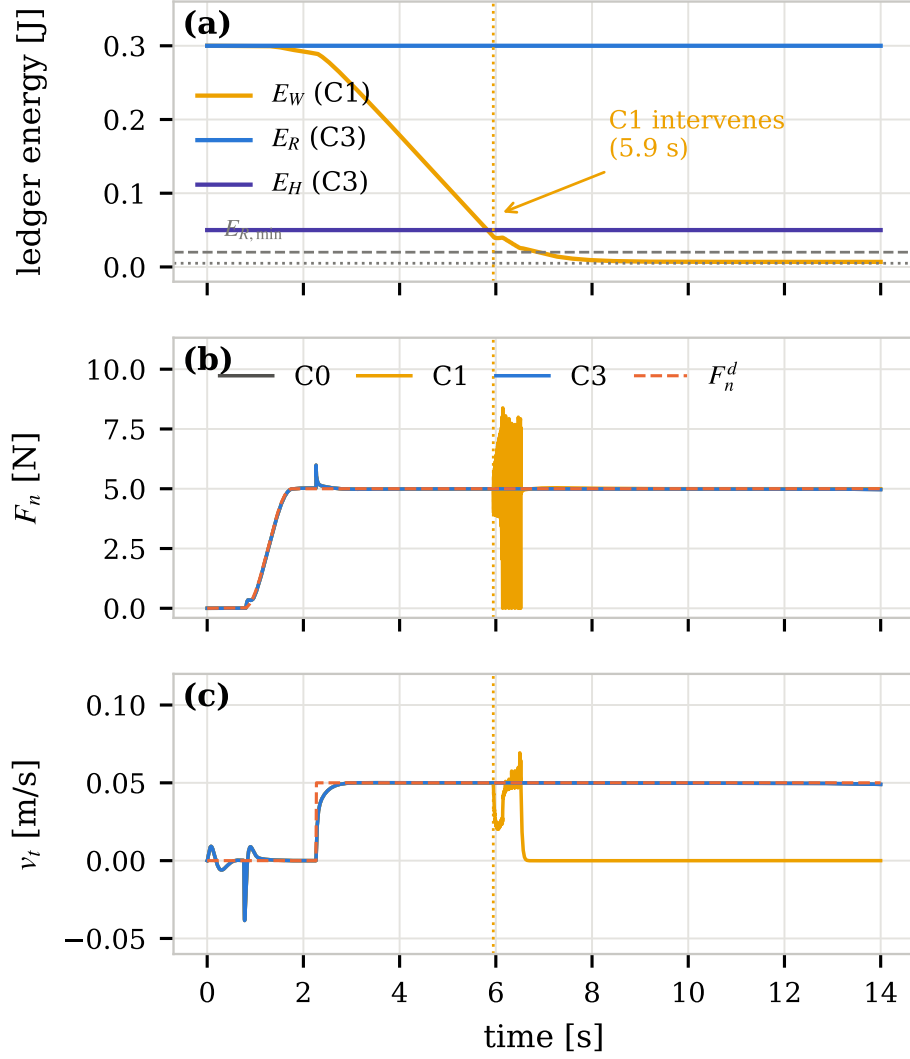


Figure 3: Exact model, no human. C1 spends its budget on ordinary friction and stops sliding. C3 leaves predictable task work unpenalized and follows the nominal force and velocity.

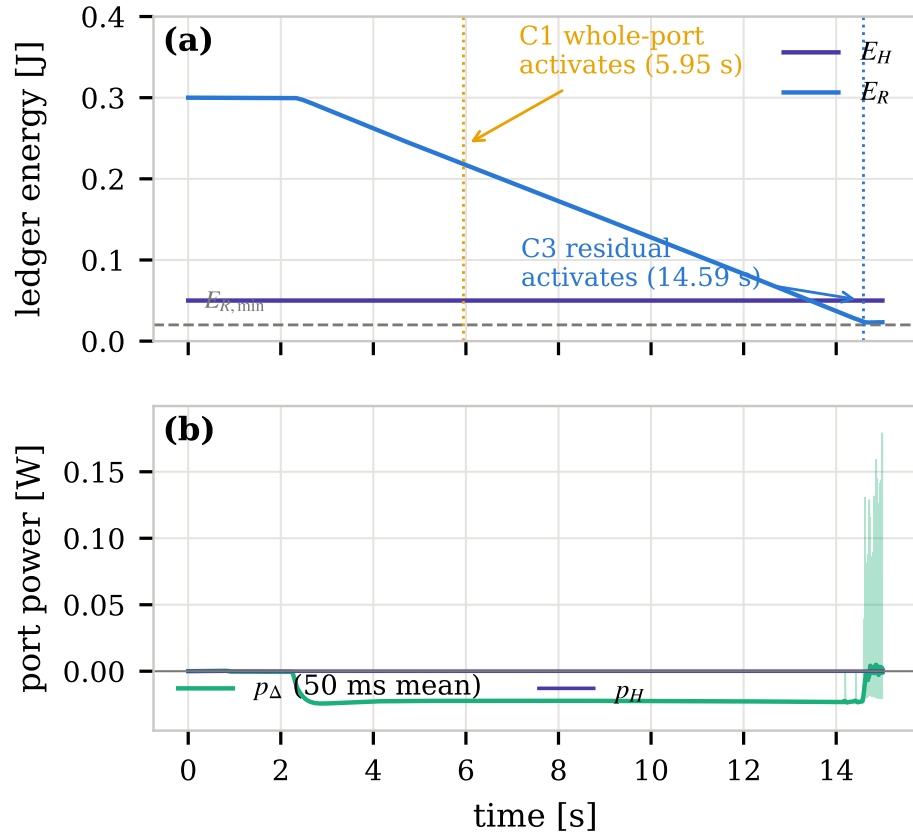


Figure 4: Friction-model mismatch without a human. Only the residual ledger depletes; the human ledger remains unchanged.

2.2 A dedicated human port limits transferred energy

Under a 3-N blocking force, C0 transfers 0.361 J to the human port and C2 transfers 0.278 J. C2 improves on C0 only after its comparatively large residual budget is exhausted, because the residual combines human input and model error. C3 separates these effects. Its instantaneous limit holds the peak human-directed power to 0.0997 W, and its 0.045-J usable human budget limits cumulative output to 0.044 J. This is an $8.3\times$ reduction relative to C0 and a $6.4\times$ reduction relative to C2. The raw human ledger remains above its 0.005-J floor (Fig. 5).

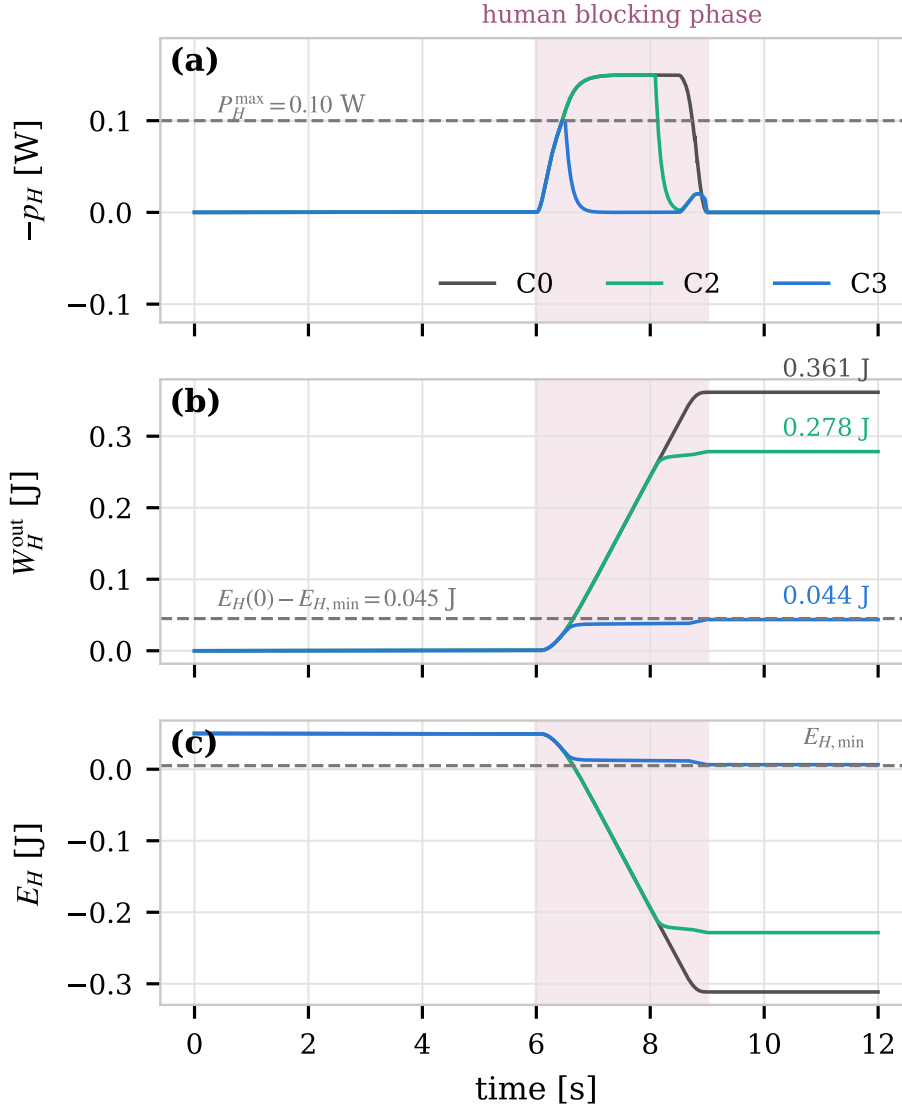


Figure 5: Blocking interaction. C3 respects both the 0.10-W instantaneous power limit and the 0.045-J human-energy budget; C0 and C2 do not enforce the dedicated human-port limits.

The two-ledger structure remains interpretable when model error and human interaction occur together. The human ledger tracks only p_H , while the residual ledger tracks $p_H + p_\Delta$ (Fig. 6). C3 remains feasible because the smaller human budget acts first; the residual-only C2 exhausts its shared budget in this combined case.

The human ledger is also bidirectional. Positive work from a helping human recharges it up to

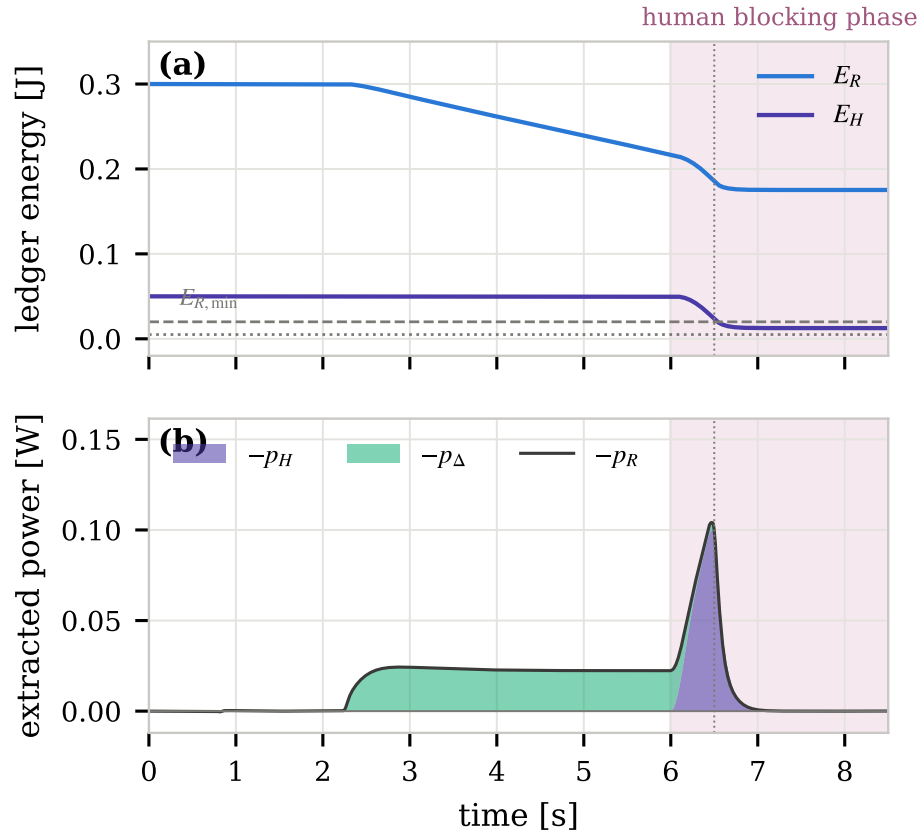


Figure 6: C3 (no governor) under simultaneous model mismatch and blocking. The human and mismatch contributions remain separated, while their sum is recorded by the residual ledger.

the 0.08-J cap; a later blocking interaction spends that stored energy before approaching the floor (Fig. 7). This behavior gives the ledger a direct energy interpretation rather than the meaning of a one-way timeout.

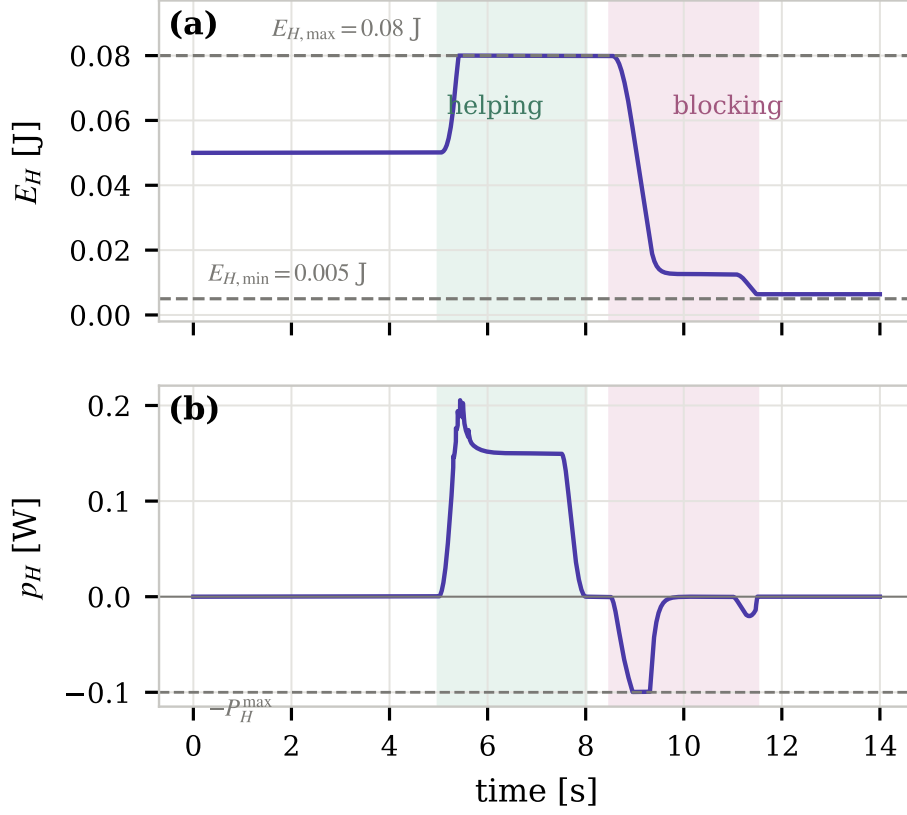


Figure 7: C3 (no governor). A helping interaction charges the human ledger; a later blocking interaction uses the stored energy while respecting the power limit.

2.3 C4 removes the recovery jump

In C3, the passivity filter can hold the robot while the moving reference continues to advance. Release then exposes the accumulated position error and causes a catch-up transient. C4 uses the same QP and safety certificate, but adds a reference governor upstream. When the human-energy constraint activates, it smoothly stops the governed reference and fixes it near the robot. After release, the reference resumes smoothly from this new position instead of reconnecting to the old time-indexed trajectory.

The fixed anchor retains position stiffness and limits yielding to 1.7 mm. The human ledger reaches a minimum of 0.0131 J, above its certified floor. The QP therefore provides the safety certificate in both C3 and C4, while the governor in C4 shapes only the interaction and recovery.

Figure 8 compares C3 and C4 over the same interaction. At release, C3 has 105.5 mm of reference debt and reaches 24.9 N command, 0.681 m/s velocity, and 19.2 N normal-force error. C4 reduces these to 7.5 mm, 3.00 N, 0.029 m/s, and 0.077 N, respectively.

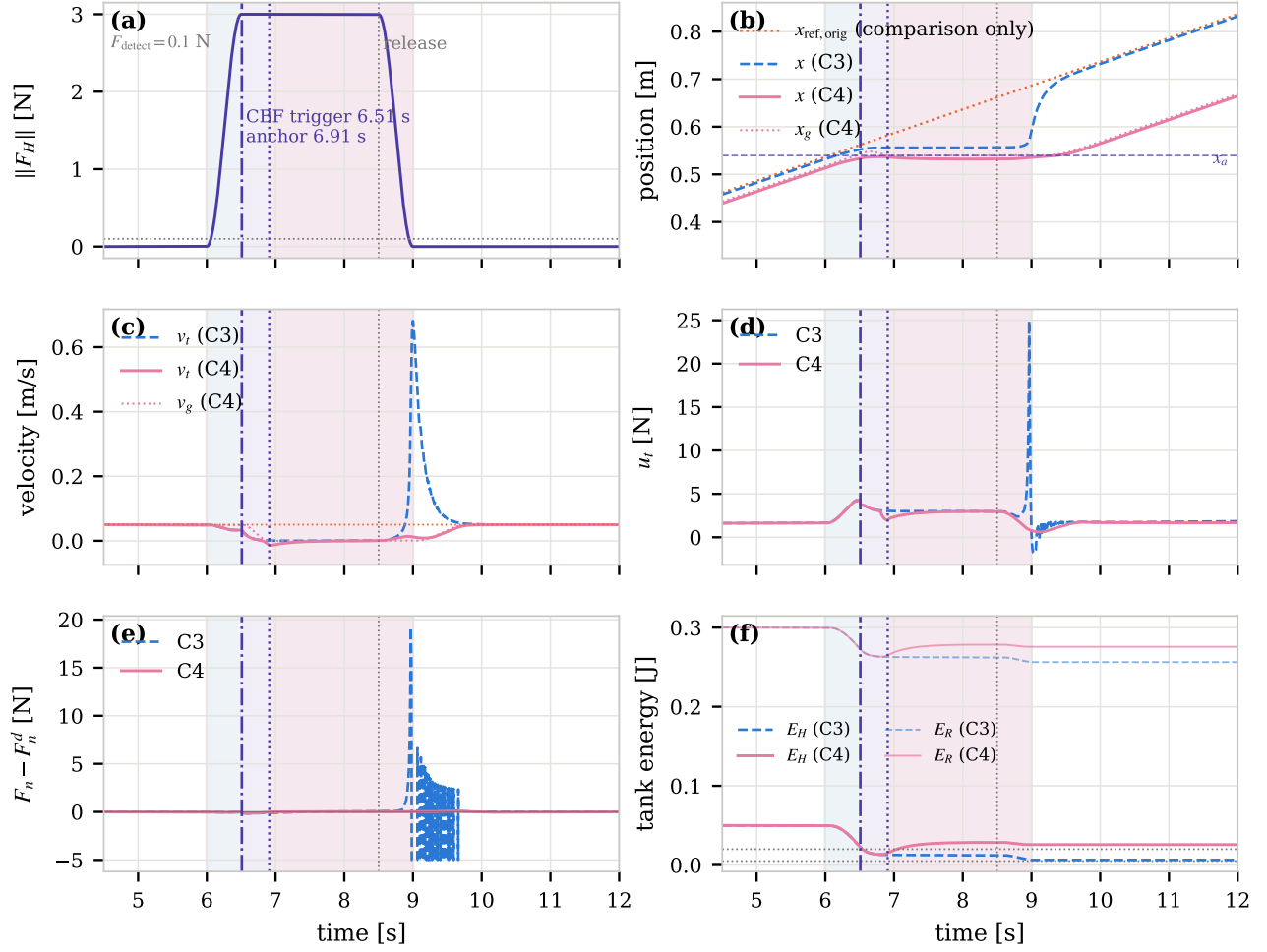


Figure 8: C3 (no governor) versus C4 (with governor) over the complete interaction and release. C4 resumes from a fixed reference anchor and avoids C3’s command and velocity burst. Both use the same passivity QP.

3 Conclusion

The PoC supports three conclusions. First, subtracting predicted task contact prevents useful work from unnecessarily consuming a passivity budget. Second, a dedicated human ledger independently bounds power and energy transferred through the human port. Third, adding the governor to C3 gives C4 a smooth release without changing the certificate. All reported C3 and C4 runs remained feasible and respected their energy floors and prediction bound. These simulation results assume known human force; physical experiments and practical force estimation remain future work.